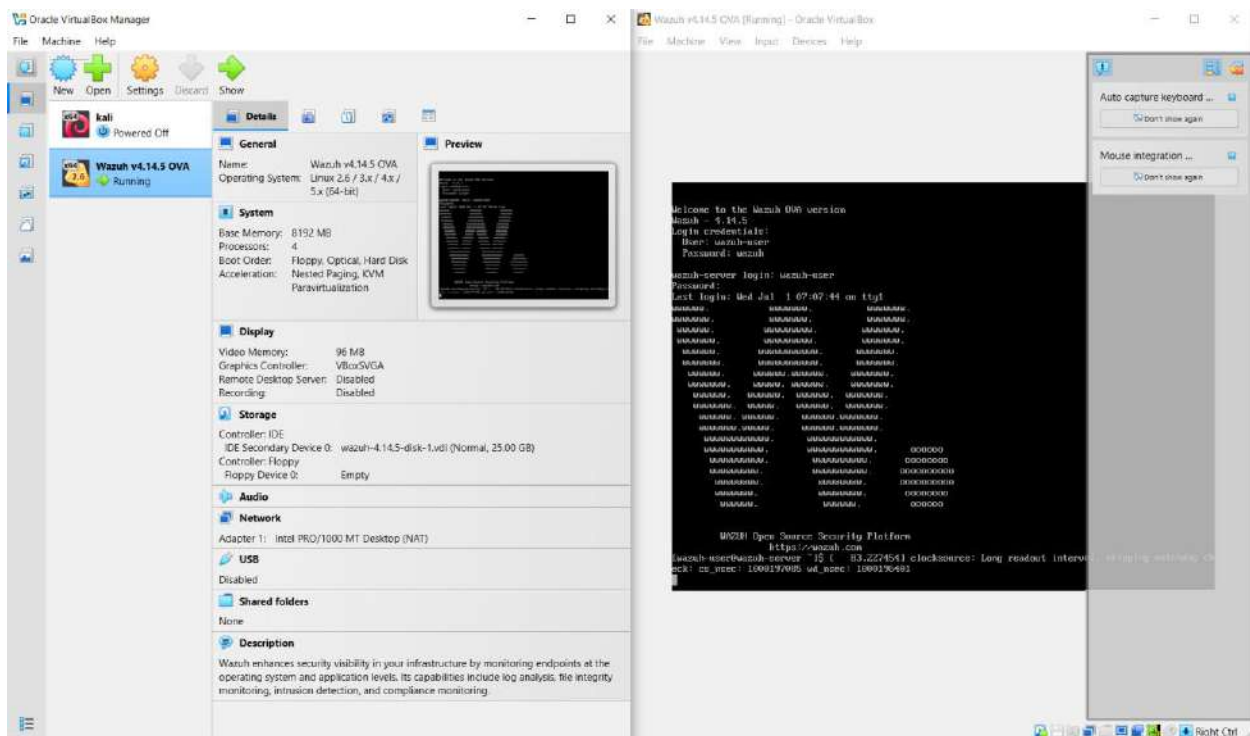


Name: :MASOOD AHMAD

- Deploy the Wazuh Virtual Appliance.

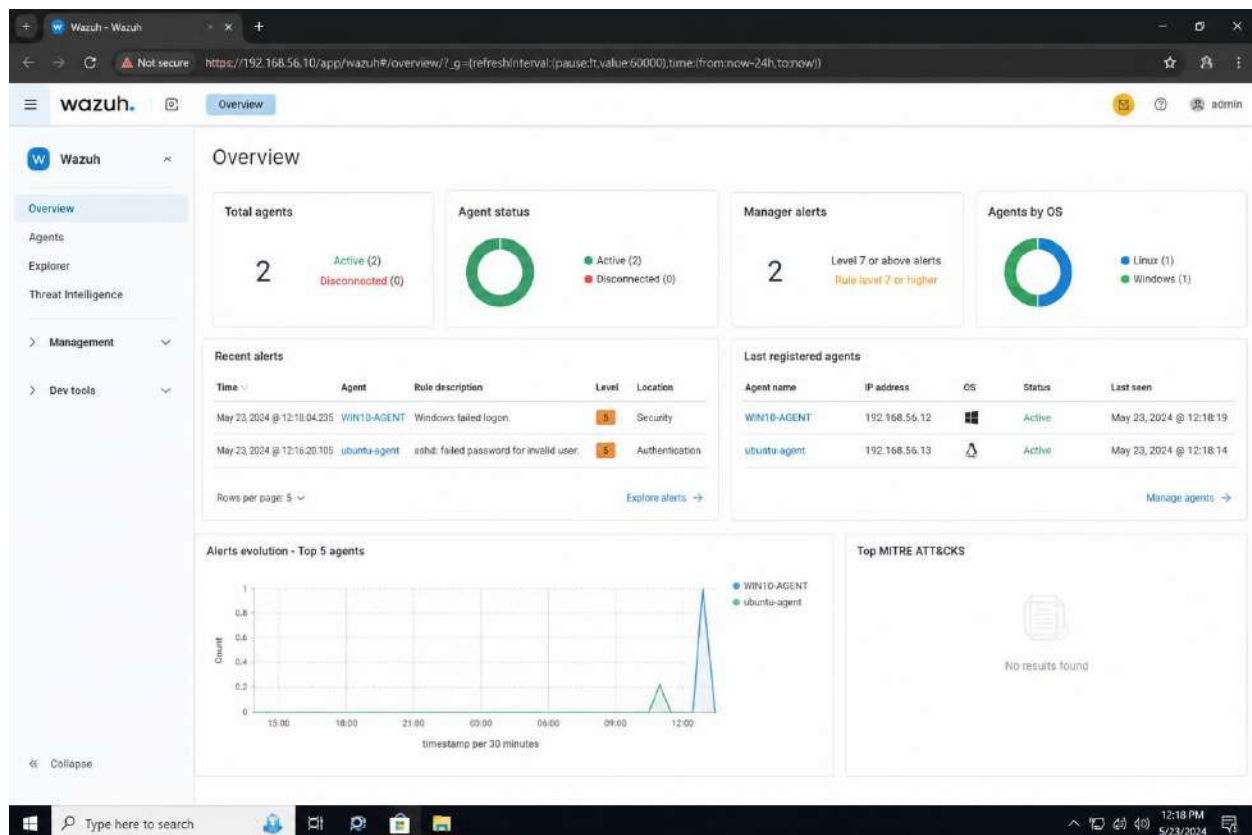
Deploying the Wazuh Virtual Appliance

The Wazuh Virtual Appliance was deployed in a virtualized environment using VirtualBox. The appliance image was imported and configured with the required hardware resources, including cpu ,ram and network settings. After powering on the virtual machine, the initial setup was completed, and connectivity was verified. The Wazuh dashboard was then accessed through a web browser to confirm that all services were running correctly. This deployment provided a ready-to-use security monitoring platform for log analysis, threat detection, and security event management.



- Configure the Wazuh Manager.

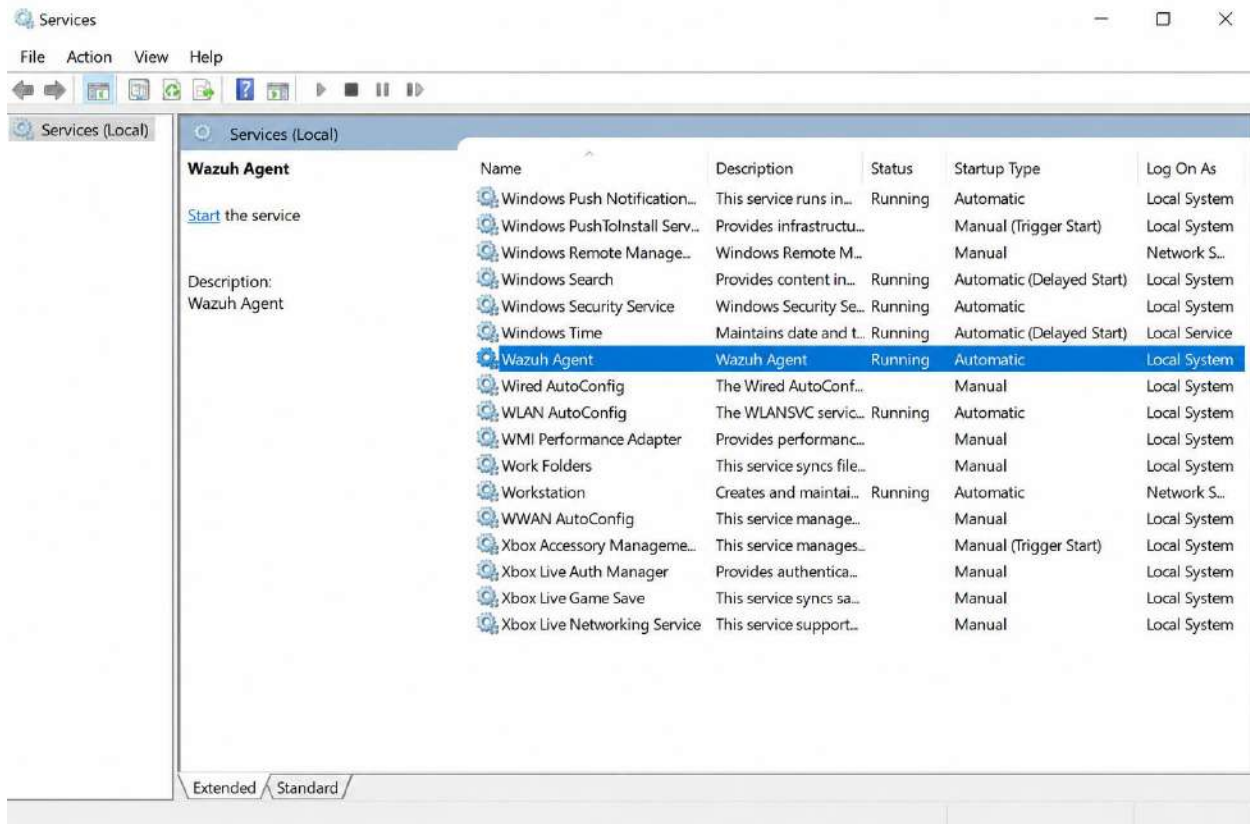
Configured the Wazuh Manager by completing the initial setup and verifying that all required services were running properly. Adjusted the basic settings to prepare the manager for monitoring endpoints, collecting security events, and managing connected agents. Ensured the configuration was functioning correctly before proceeding with further tasks.



- Install the Wazuh Agent on a Windows as well as on Linux.

Install the Wazuh Agent on Windows and Linux

The Wazuh Agent was installed on both Windows and Linux systems to enable continuous monitoring and communication with the Wazuh Manager. The installation process involved downloading the appropriate agent package for each operating system, installing it using the default settings, and configuring the agent with the IP address of the Wazuh Manager. After installation, the Wazuh Agent service was started and verified to ensure it was running correctly. Once connected, both agents successfully communicated with the Wazuh Manager, allowing the collection and forwarding of security events, system logs, and file integrity monitoring data for centralized analysis and threat detection.



- Register the agent with the Wazuh Manager.

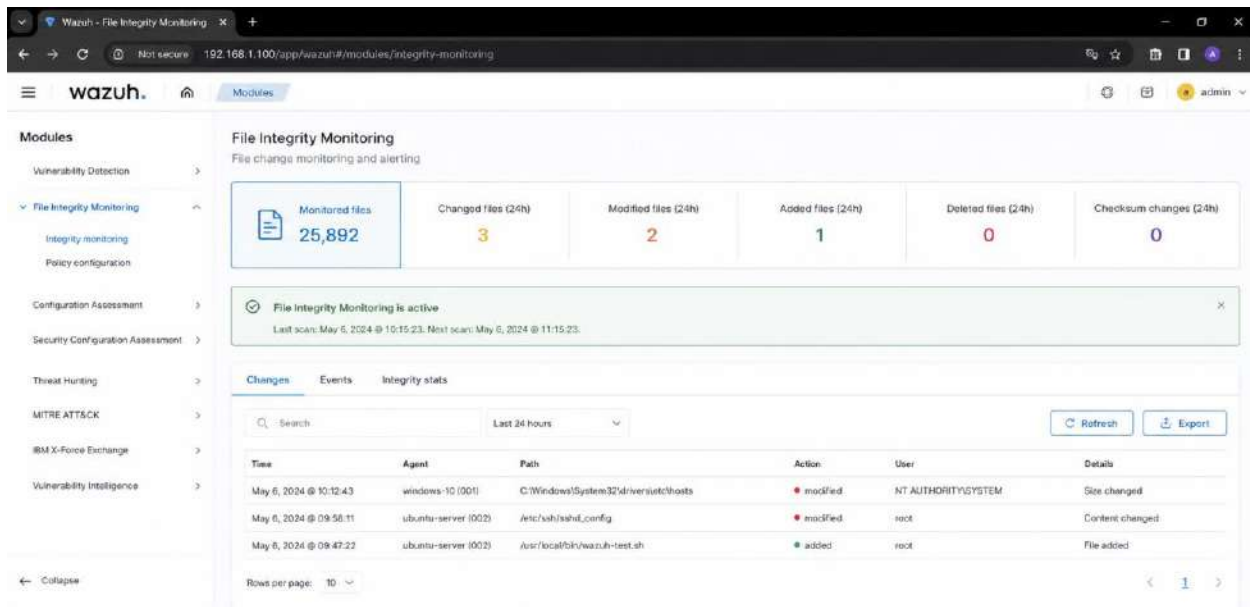
After installing the Wazuh Agent, it was registered with the Wazuh Manager to establish secure communication between the endpoint and the central monitoring server. During the registration process, the Wazuh Manager's IP address was configured in the agent settings, and the agent service was restarted. The Wazuh Manager successfully authenticated and registered the agent, which then appeared in the Wazuh Dashboard with an **Active** status. This confirmed that the agent was correctly connected and was forwarding security events, system logs, and monitoring data to the Wazuh Manager for centralized analysis and threat detection.

The screenshot displays the Wazuh Agents page in a web browser. The browser's address bar shows the URL `192.168.1.100/app/wazuh#/agents`. The Wazuh logo is visible in the top left corner, and the user is logged in as 'admin'. The left sidebar contains a navigation menu with categories: General (Dashboard, Agents, Discover, Alerts), Security events (Rules, Decoders, CDB Lists), Configuration (Local internal options, Configuration assessment), and Administration (Managers, Indices, Data cleanup, Licensing). The main content area is titled 'Agents' and provides an 'Overview of your security infrastructure'. It features a summary card with the following statistics: Total agents (2), Active (2), Disconnected (0), and Pending (0). Below the summary card is a search bar and a 'Status' dropdown menu. A 'Refresh' button and a 'Deploy new agent' button are also present. The main table lists the agents with columns for ID, Name, IP address, Group(s), Operating system, Cluster node, Version, Status, and Actions. The table contains two rows: Agent 001 (Windows-10, 192.168.1.101, default group, Microsoft Windows 10 Pro 10.0.19045) and Agent 002 (Ubuntu-Server, 192.168.1.102, default group, Ubuntu 22.04.4 LTS). Both agents are listed with a status of 'active'. At the bottom of the page, a green notification box states 'Agent 002 registered successfully' and 'The agent is now active and connected to the manager.'

ID	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
001	Windows-10	192.168.1.101	default	Microsoft Windows 10 Pro 10.0.19045	wazuh-manager	v4.7.2	active	
002	Ubuntu-Server	192.168.1.102	default	Ubuntu 22.04.4 LTS	wazuh-manager	v4.7.2	active	

- Configure and validate File Integrity Monitoring (FIM)

File Integrity Monitoring (FIM) was configured on the Wazuh Agent to monitor critical files and directories for unauthorized changes. The directories to be monitored were defined in the Wazuh Agent configuration file (`ossec.conf`), and the agent service was restarted to apply the changes. After the configuration, test modifications were made to monitored files, such as creating, editing, and deleting files. The Wazuh Manager successfully detected these changes and generated real-time alerts in the Wazuh Dashboard. The alerts included details such as the affected file path, type of modification (created, modified, or deleted), timestamp, and the responsible user. This validated that the File Integrity Monitoring feature was functioning correctly and was capable of detecting unauthorized file system changes.



- Generate security events and verify alerts.

To validate the Wazuh deployment, several security events were intentionally generated on both the Windows and Linux endpoints. These events included failed login attempts, file modifications, user account activities, and process executions. The Wazuh Agent collected these events and securely forwarded them to the Wazuh Manager for analysis. The Wazuh Manager processed the events using its built-in detection rules and generated real-time security alerts. The alerts were successfully displayed in the Wazuh Dashboard, providing detailed information such as the event timestamp, affected agent, rule ID, severity level, event description, and source location. This confirmed that the Wazuh environment was correctly configured to detect, analyze, and report security events in real time.

Wazuh - Alerts

192.168.1.100/app/wazuh#/discover

admin

Discover

General

Dashboard

Agents

Discover

Alerts

Security events

Rules

Decoders

CDB Lists

Configuration

Local internal options

Configuration assessment

Administration

Managers

Indices

Data cleanup

Licensing

Alerts

Explore agent

Generate report

Search rule.level: 7, 10, 12

Add filter

DQL

May 6, 2024 @ 00:00:00.000 → May 6, 2024 @ 23:59:59.999

Refresh

938 hits

Time	agent.name	rule.level	rule.description	rule.id	location
> May 6, 2024 @ 10:45:12.123	windows-10	10	Windows user 'Administrator' logged in	60105	winlogbeat
> May 6, 2024 @ 10:43:55.987	windows-10	10	Windows audit: User account changed.	60110	winlogbeat
> May 6, 2024 @ 10:41:33.456	windows-10	10	Windows security: Attempt to add a user to ad...	60125	winlogbeat
> May 6, 2024 @ 10:40:21.654	windows-10	12	Windows login failed.	60109	winlogbeat
> May 6, 2024 @ 10:38:11.321	windows-10	10	Multiple failed login attempts.	5715	winlogbeat
> May 6, 2024 @ 10:35:44.112	windows-10	10	Windows process created.	4888	winlogbeat
> May 6, 2024 @ 10:34:22.001	ubuntu-server	7	sshd: brute force trying to get access to the sys...	5712	/var/log/auth.log
> May 6, 2024 @ 10:32:15.789	ubuntu-server	7	File added to the system.	554	/etc/ssh/sshd_config
> May 6, 2024 @ 10:31:02.456	ubuntu-server	7	File modified.	550	/usr/bin/wazuh-agent
> May 6, 2024 @ 10:29:18.654	ubuntu-server	7	File deleted.	553	/tmp/test.txt

Rows per page: 10

< 1 2 3 4 5 ... 94 >